

PROJECT REPORT

# **Fraud Detection in Financial Transactions Using Machine Learning**

A Machine Learning Approach to Identifying Fraudulent Digital Payments

**Prepared by:** Nirbhavane Vedant Prakash

**Institution:** Matoshri Asarabai Institute of Technology and Research Centre

**Academic Year:**

2025-26

# Certificate

This is to certify that the project report entitled **“Fraud Detection in Financial Transactions Using Machine Learning”** is a bona fide work carried out by the student named on the title page under the prescribed academic requirements. The work presented in this report has not been submitted elsewhere for the award of any degree or diploma.

Date: \_\_\_\_\_ Place: \_\_\_\_\_

## Declaration

I hereby declare that this project report is my original work. The ideas, analyses, and results presented here have been prepared for academic purposes, and all sources consulted have been acknowledged in the references.

## Acknowledgement

I express my sincere gratitude to my project guide, faculty members, and institution for their guidance and support. I also thank the developers of the Python data-science ecosystem and the creators of the open documentation used during this project

# Abstract

Financial fraud is a major challenge in digital payment systems, where fraudulent transactions can cause significant financial losses. This project develops a machine learning-based system for detecting fraudulent financial transactions. The dataset was analyzed and preprocessed by handling categorical variables, preparing numerical features, and separating the target variable. Exploratory data analysis was performed to understand transaction patterns and identify important features.

Logistic Regression, Decision Tree, and Random Forest models were evaluated. Random Forest was selected as the final model because of its stronger ability to distinguish normal and fraudulent transactions. Probability threshold tuning was applied to improve the precision-recall balance. The final model achieved 99.97% accuracy, 92.80% precision, 80.77% recall, 86.37% F1-score, and approximately 99.90% ROC-AUC. These results show that machine learning can provide a strong foundation for automated fraud detection systems.

# Table of Contents

Chapter 1 - Introduction

Chapter 2 - Literature Review

Chapter 3 - Dataset & Data  
Preprocessing

Chapter 4 - Exploratory Data Analysis

Chapter 5 - Methodology

Chapter 6 - Machine Learning Models

Chapter 7 - Model Evaluation & Results

Chapter 8 - Feature Importance &  
Analysis

Chapter 9 - Conclusion & Future Scope

References

Appendix - Important Code

## List of Figures

Figure 1. Initial model performance comparison

Figure 2. Random Forest confusion matrix

Figure 3. ROC curve - Random Forest

Figure 4. Random Forest feature importance

## List of Tables

Table 1. Dataset features

Table 2. Random Forest configuration

Table 3. Model comparison

Table 4. Final model resul

# Chapter 1 - Introduction

## 1.1 Background

The rapid growth of digital payments and online financial transactions has increased the importance of transaction security. Fraudulent transactions can lead to financial losses and harm both customers and

financial institutions. Traditional rule-based systems may miss complex and evolving patterns; machine learning learns from historical data to flag transactions likely to be fraudulent.

## 1.2 Problem Statement

To develop a machine learning model capable of distinguishing fraudulent financial transactions from normal transactions and to evaluate it using appropriate classification metrics.

## 1.3 Objectives

- Analyze financial transaction data.
- Perform preprocessing and exploratory data analysis.
- Identify features related to fraudulent transactions.
- Train and compare multiple classification models.
- Select the best model and improve it through threshold tuning.
- Evaluate Accuracy, Precision, Recall, F1-score and ROC-AUC.

## 1.4 Scope

The project focuses on binary classification: Normal (0) and Fraud (1). The resulting model can serve as a foundation for automated fraud detection in digital payment environments.

## Chapter 2 - Literature Review

### 2.1 Related Work

Machine learning is widely used in financial fraud detection because it can identify patterns in large transaction datasets. Classification algorithms such as Logistic Regression, Decision Tree, and Random Forest are commonly used to separate normal behavior from fraudulent activity.

Logistic Regression supplies a simple and interpretable baseline for binary classification, although its performance can be limited when the relationship between transaction attributes and fraud is complex.

Decision Trees capture nonlinear relationships and are easy to interpret, but a single tree can overfit the training data. Random Forest combines many decision trees, generally improving generalization and robustness when features interact in nonlinear ways.

Fraud datasets are usually highly imbalanced. Accuracy alone can therefore be misleading; precision, recall, F1-score, and ROC-AUC are essential for evaluating whether a model detects fraud without generating an excessive number of false alerts.



## Chapter 3 - Dataset & Data Preprocessing

### 3.1 Dataset

The dataset contains financial transaction records with transaction type, amount, sender and receiver balances, time step, and fraud status. The target variable is **isFraud**: 0 represents a normal transaction and 1 represents a fraudulent transaction.

### 3.2 Important Features

| Feature        | Description                        |
|----------------|------------------------------------|
| step           | Time step of the transaction       |
| type           | Type of transaction                |
| amount         | Transaction amount                 |
| oldbalanceOrg  | Original sender balance            |
| newbalanceOrig | Sender balance after transaction   |
| oldbalanceDest | Original receiver balance          |
| newbalanceDest | Receiver balance after transaction |
| isFraud        | Fraud target variable              |

### 3.3 Data Preparation

The dataset was examined for missing values, duplicate records, and unsuitable data types. Categorical values were encoded numerically, numerical features were prepared for modeling, and the target variable was separated from the predictor variables.

# Chapter 4 - Exploratory Data Analysis

## 4.1 Analysis Approach

Exploratory data analysis was used to understand transaction distributions, relationships between account balances and transaction amounts, and the class imbalance between normal and fraudulent transactions. Such examination is vital because a very small fraud class can make a seemingly accurate model operationally weak.

## 4.2 Observations

Balance-related features and transaction amount were expected to carry meaningful predictive information. The model's later feature-importance scores confirmed that sender balance before a transaction, amount, and sender balance after a transaction were especially informative.

## Chapter 5 - Methodology

### 5.1 Handling Categorical Variables

The transaction **type** variable was transformed into numerical indicator variables, enabling the classification algorithms to process transaction-category information.

### 5.2 Feature Selection

Feature importance from Random Forest was analyzed after modeling. The most influential reported features were oldbalanceOrg (0.294996), amount (0.163399), newbalanceOrig (0.157240), and type\_TRANSFER (0.102957).

### 5.3 Train-Test Split

The data was divided into training and testing sets. Training data was used to fit each classifier, while the held-out testing data was used only to measure performance on unseen transactions.

### 5.4 Threshold Tuning

Because fraud detection is imbalanced, the default probability threshold may produce an unsuitable precision-recall tradeoff. The Random Forest probability threshold was adjusted to improve precision while retaining a reasonable recall.

# Chapter 6 - Machine Learning Models

## 6.1 Evaluated Models

**Logistic Regression.** Used as an interpretable baseline for fraud classification.

**Decision Tree.** Implemented to capture nonlinear feature relationships.

**Random Forest.** An ensemble of decision trees selected as the final approach because of its strong overall classification performance.

## 6.2 Random Forest Configuration

| Parameter            | Value         |
|----------------------|---------------|
| Algorithm            | Random Forest |
| Number of estimators | 100           |
| Maximum depth        | 12            |
| Class weight         | Balanced      |
| Random state         | 42            |

The model was trained using the training dataset and evaluated on the testing dataset. Balanced class weights were used to help account for the rare fraud class.

# Chapter 7 - Model Evaluation & Results

## 7.1 Evaluation Metrics

Accuracy measures the share of correct classifications. Precision measures how many fraud alerts are truly fraudulent. Recall measures how many actual fraud cases are detected. F1-score balances precision and recall. ROC-AUC measures discrimination ability across thresholds.

## 7.2 Initial Model Comparison

| Model               | Precision | Recall | F1-Score | ROC-AUC |
|---------------------|-----------|--------|----------|---------|
| Logistic Regression | 0.0288    | 0.9404 | 0.0559   | 0.9888  |
| Decision Tree       | 0.1433    | 0.9872 | 0.2503   | 0.9925  |
| Random Forest       | 0.1203    | 0.9866 | 0.2144   | 0.9990  |

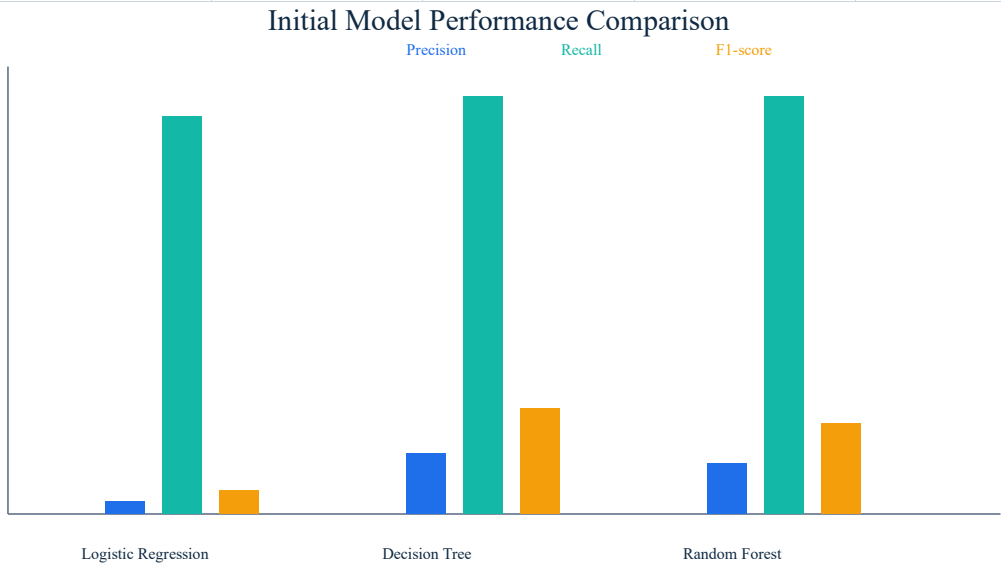


Figure 1. Initial model comparison. Threshold tuning subsequently improved the Random Forest precision-recall balance.

## 7.3 Final Random Forest Results

| Metric    | Result |
|-----------|--------|
| Accuracy  | 99.97% |
| Precision | 92.80% |
| Recall    | 80.77% |
| F1-Score  | 86.37% |
| ROC-AUC   | 99.90% |

Random Forest Confusion Matrix

|        |        | Predicted |        |
|--------|--------|-----------|--------|
|        |        | Fraud     | Normal |
| Actual | Fraud  | 1,270,778 | 103    |
|        | Normal | 316       | 1,327  |

Figure 2. Final Random Forest confusion matrix.

The final confusion matrix shows 1,270,778 normal transactions correctly identified as normal and 1,327 fraudulent transactions correctly identified. It also records 103 false positives and 316 false negatives.

# Chapter 8 - Feature Importance & Analysis

## 8.1 Feature Importance

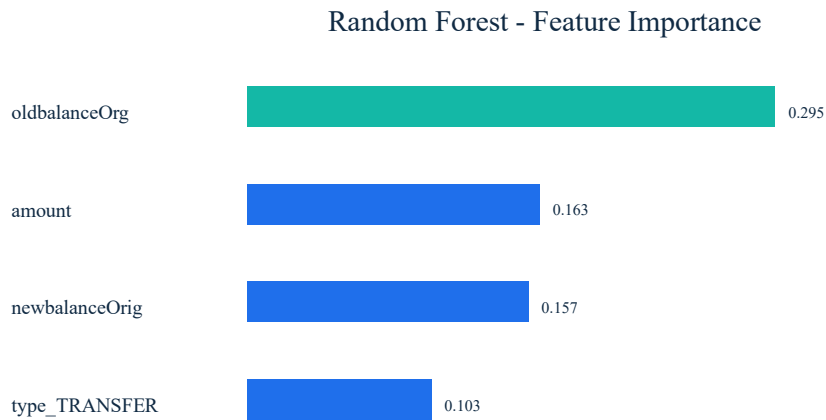


Figure 3. Random Forest feature importance.

Feature importance analysis indicates that oldbalanceOrg was the most influential feature, followed by amount and newbalanceOrig. These variables capture account-balance changes and transaction values that help distinguish potentially fraudulent activity.

## 8.2 ROC Curve

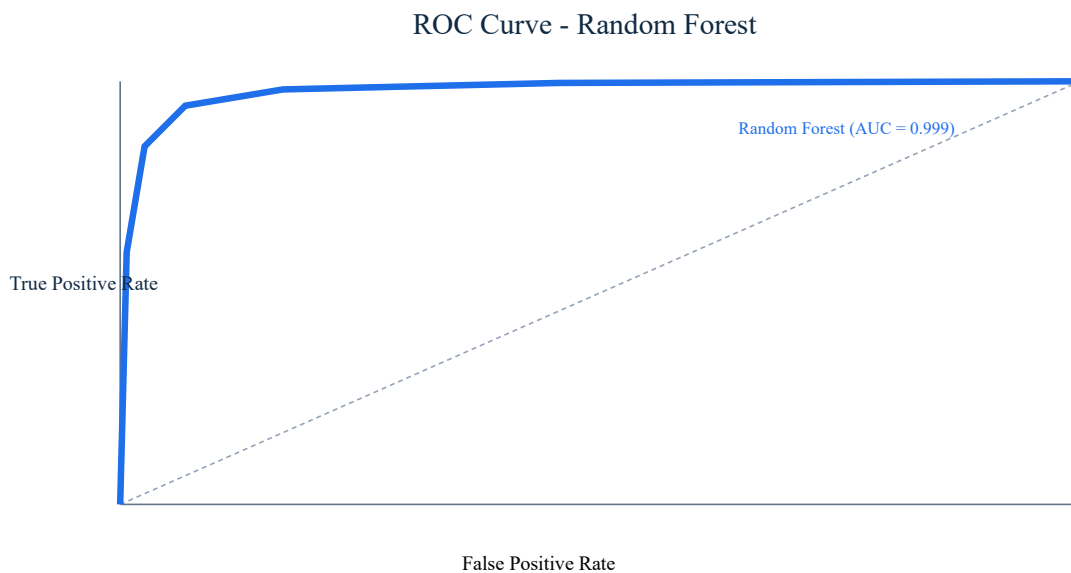


Figure 4. ROC curve - Random Forest.

The reported ROC-AUC of approximately 0.999 indicates excellent discrimination between normal and fraudulent transactions across classification thresholds.

# Chapter 9 - Conclusion & Future Scope

## 9.1 Conclusion

This project successfully developed a machine learning-based fraud detection system for identifying fraudulent financial transactions. Multiple classification algorithms were evaluated, including Logistic Regression, Decision Tree and Random Forest.

Random Forest was selected as the final model because of its strong performance. After threshold tuning, the model achieved 99.97% accuracy, 92.80% precision, 80.77% recall, 86.37% F1-score and approximately 99.90% ROC-AUC.

Feature importance analysis showed that account-balance-related features and transaction amount played an important role in fraud detection. Overall, the results demonstrate that machine learning can be effectively applied to fraud detection and can provide a strong foundation for automated fraud monitoring systems.

## 9.2 Advantages

- High fraud detection performance.
- Handles nonlinear relationships.
- Relatively robust to noise.
- Provides feature importance.
- Threshold tuning adjusts precision and recall.
- Can process large transaction volumes efficiently.

## 9.3 Limitations

- The dataset may not represent every real-world fraud pattern.
- Fraud patterns can change over time.
- Highly imbalanced data makes evaluation challenging.
- False negatives can still occur.
- The model needs retraining when patterns shift.

## 9.4 Future Scope

Future work can implement real-time detection with streaming transaction data, hyperparameter optimization, explainable AI, anomaly-detection methods, and continuous retraining using newly observed transactions.



## References

Scikit-learn documentation - Machine learning algorithms and evaluation metrics.

Python documentation.

Pandas documentation.

NumPy documentation.

Matplotlib documentation.

Seaborn documentation.

Research papers related to machine learning-based financial fraud detection.

## Appendix - Important Code

The following pseudocode summarizes the essential modeling workflow. The original notebook code was not included with the supplied report content.

| Step                  | Illustrative operation                                                                                        |
|-----------------------|---------------------------------------------------------------------------------------------------------------|
| Prepare features      | Encode transaction type; retain numerical balance and amount fields.                                          |
| Split data            | Create training and testing subsets.                                                                          |
| Train model           | <code>RandomForestClassifier(n_estimators=100, max_depth=12, class_weight="balanced", random_state=42)</code> |
| Predict probabilities | Generate fraud probability for each test transaction.                                                         |
| Tune threshold        | Convert probabilities to labels using the chosen probability threshold.                                       |
| Evaluate              | Calculate precision, recall, F1-score, ROC-AUC, and confusion matrix.                                         |